

# Touchdot S3 Development Board Product Brief

Compact ESP32-S3 mini microcontroller development board inspired by Lilypad. Ideal for IoT, control systems, and creative electronics projects.

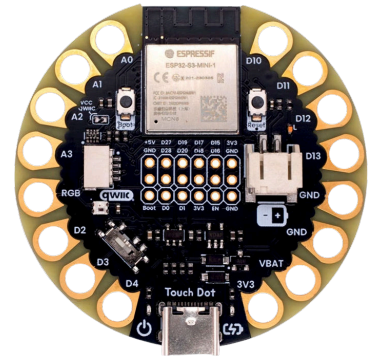
Version: 1.0

Modified: 2025-12-09 08:59

## Introduction

The Unit Touchdot S3 is a compact ESP32-S3 Mini board designed for wearable electronics, IoT devices, educational projects, and smart automation. Its Lilypad-inspired, low-profile, sewable design makes it perfect for integrating into textiles and compact enclosures while providing advanced wireless and processing capabilities.

Equipped with 2.4 GHz Wi-Fi, Bluetooth 5.0, and a modern interface featuring a USB-C connector, onboard LiPo charging, and a QWIIC I<sup>2</sup>C port, the board supports rapid prototyping and creative development. It offers a robust platform bridging wearable design and embedded computing, tailored for students, makers, and engineers.



## Settings

### Interface Overview

| Interface | Signals / Pins | Typical Use                                       |
|-----------|----------------|---------------------------------------------------|
| UART      | TX (GPIO17)    | Serial transmit (TX)                              |
| UART      | RX (GPIO16)    | Serial receive (RX)                               |
| I2C       | SDA (GPIO5)    | I <sup>2</sup> C data line (QWIIC, OLED, sensors) |
| I2C       | SCL (GPIO6)    | I <sup>2</sup> C clock line                       |
| SPI       | MOSI (GPIO9)   | Data to SPI device                                |
| SPI       | MISO (GPIO8)   | Data from SPI device                              |
| SPI       | SCK (GPIO13)   | SPI clock signal                                  |
| SPI       | CS (GPIO21)    | Chip select for SPI device                        |
| USB       | D+ (GPIO20)    | USB differential data (+)                         |
| USB       | D (GPIO19)     | USB differential data (-)                         |

### Supported Pins

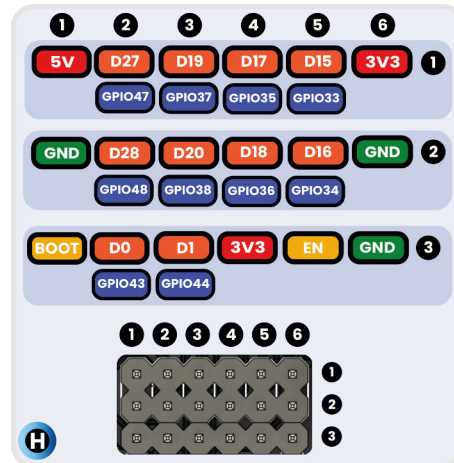
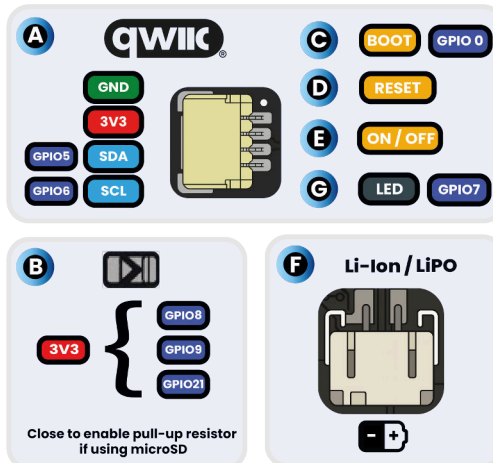
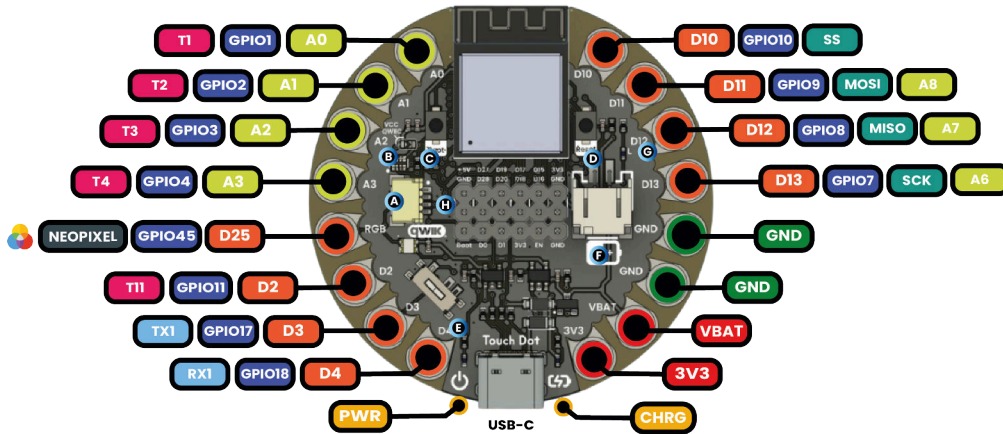
| Symbol      | I/O           | Description                                    |
|-------------|---------------|------------------------------------------------|
| USB -C      | Input         | USB-C connector for 5V power and data          |
| Li-ion/LiPo | Input         | Connector for LiPo battery power (3.7V - 4.2V) |
| VCC         | Input         | Main power supply (3.3V)                       |
| GND         | Ground        | Ground connection                              |
| IO          | Bidirectional | General-purpose I/O pins                       |
| NeoPixel    | Output GPIO25 | WS2812 RGB LED data output                     |

### Pin & Connector Layout

| Group    | Available pins | Suggested use                          |
|----------|----------------|----------------------------------------|
| GPIO     | D2 to D13      | Sensors, actuators                     |
| UART     | Tx and Rx      | Serial communication                   |
| TouchPad | T1 to T11      | Capacitive sensors for touch detection |
| Analog   | A0 to A8       | 12-bit (0–4095) resolution             |
| SPI      | Optional       | Displays, additional memory            |

## Block Diagram

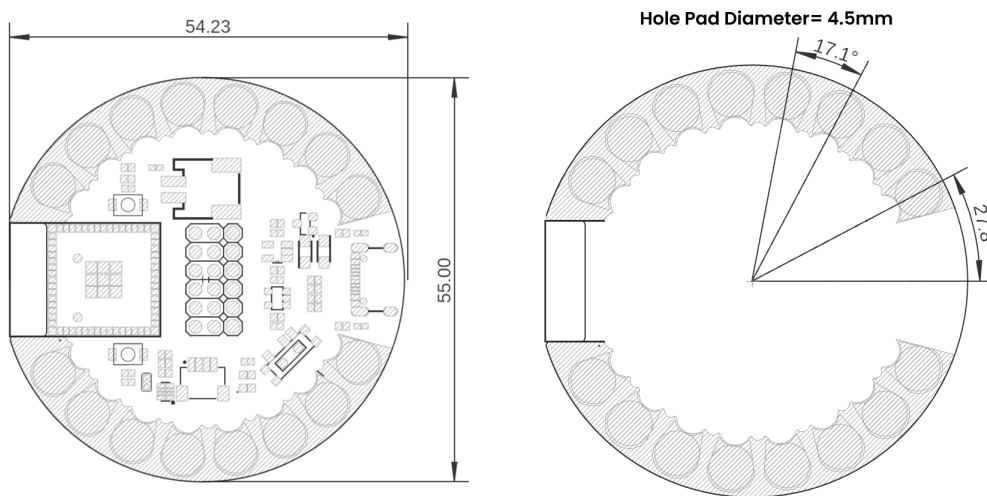
# Top view



## Description:

- Supply voltage
- GND
- Components
- GPIO compatible with Arduino
- GPIO ESP32
- UART
- Touch pins
- Analog pins
- SPI
- I2C
- RGB

## Dimensions



Mechanical dimensions in millimeters of Touch Dot S3

## Usage

- Arduino IDE (ESP32 board manager)
- ESP-IDF toolchain
- MicroPython firmware
- CircuitPython (via UF2 bootloader)

## Downloads

- Schematic PDF
- Board Dimensions
- Pinout Diagram PNG

## Purchase

- Buy from UNIT Electronics
- Open product page